

CPEN 416 / EECE 5710

Lecture 15: Order finding and Shor's algorithm

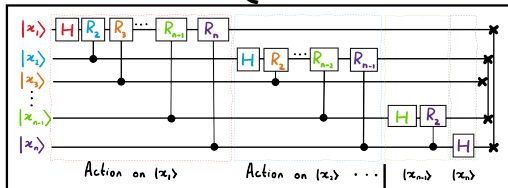
Thursday 30 October 2025

Announcements

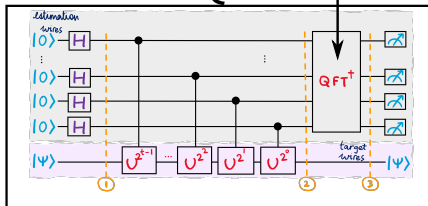
- Tutorial Assignment 3 due tomorrow 23:59
- Midterm checkpoint report due **tomorrow by 5:00pm**
 - Meeting appointments open in Canvas later this afternoon
- Tutorial 7 Monday: RSA and Shor's algorithm
- Quiz 7 Tuesday

Reminder: where are we going?

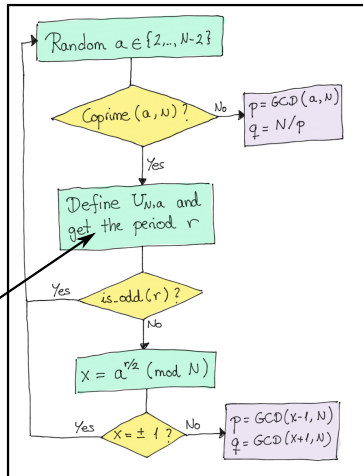
1. QFT



2. QPE



3. Shor



Last time

Input: integers a and N , where $N \geq 2$ and $\gcd(a, N) = 1$

Output: smallest integer r for which $a^r \equiv 1 \pmod{N}$

Exercise: find the order of $a = 8$ for $N = 15$.

$$8^r \equiv 1 \pmod{15}$$

$\Rightarrow r = 4$

$$8^2 = 64 \equiv 4 \pmod{15}$$
$$8^3 = 32 \equiv 2 \pmod{15}$$
$$8^3 = \underbrace{8^2}_{4} \cdot 8$$

Learning outcomes

- Apply QPE to solve the order finding problem
- Reduce order finding to integer factorization, and implement Shor's quantum algorithm in PennyLane
- Explain the relevance, and the societal and ethical implications, of Shor's algorithm

Order finding on a quantum computer

$$N=15, a=8 \quad U_{15,8} \underset{13}{|0011\rangle} = \underset{19}{|1001\rangle}$$

Given a, N with $\gcd(a, N) = 1$, consider the quantum operation for multiplication by a modulo N ,

$$U_{N,a} |k\rangle = |ak \bmod N\rangle$$

If r is the order of a modulo N , and we apply $U_{N,a}$ r times,

$$\begin{aligned} (U_{N,a})^r |k\rangle &= |k \cdot (a)^r \bmod N\rangle \\ &= |k\rangle \end{aligned}$$

Order finding on a quantum computer

$U_{N,a}$ has the same order, r . We can use QPE to efficiently find r with a clever initialization of the target register.

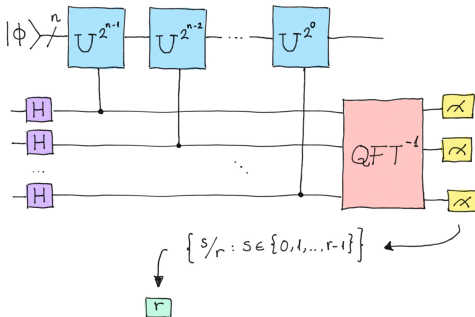


Image credit: Xanadu Quantum Codebook node S.3

Order finding on a quantum computer

Suppose there exists $|\phi\rangle$ for which

$$(U_{N,a})^r |\phi\rangle = U^r |\phi\rangle = |\phi\rangle \Rightarrow 11>$$

Consider the superposition state

$$|\psi_0\rangle = \frac{1}{\sqrt{r}} \left(|\phi\rangle + U|\phi\rangle + U^2|\phi\rangle + \dots + U^{r-1}|\phi\rangle \right)$$

Apply U to $|\psi_0\rangle$:

$$\begin{aligned} U|\psi_0\rangle &= \frac{1}{\sqrt{r}} \left(U|\phi\rangle + U^2|\phi\rangle + U^3|\phi\rangle + \dots + U^r|\phi\rangle \right) \\ &= |\psi_0\rangle \end{aligned}$$

Handwritten diagram: A red circle highlights the term $U^r|\phi\rangle$ in the equation above. A red arrow points from this circle to the $|\phi\rangle$ term in the equation $(U_{N,a})^r |\phi\rangle = U^r |\phi\rangle = |\phi\rangle$ at the top of the slide.

Order finding on a quantum computer

Now consider the state

$$|\psi_1\rangle = \frac{1}{\sqrt{r}} \left(|\phi\rangle + e^{-\frac{2\pi i}{r}} U|\phi\rangle + e^{-2 \cdot \frac{2\pi i}{r}} U^2|\phi\rangle + \dots + e^{-(r-1) \cdot \frac{2\pi i}{r}} U^{r-1}|\phi\rangle \right)$$

Apply U to $|\psi_1\rangle$:

$$\begin{aligned} U|\psi_1\rangle &= \frac{1}{\sqrt{r}} \left(U|\phi\rangle + e^{-\frac{2\pi i}{r}} U^2|\phi\rangle + e^{-2 \cdot \frac{2\pi i}{r}} U^3|\phi\rangle + \dots + e^{-(r-1) \cdot \frac{2\pi i}{r}} U^r|\phi\rangle \right) \\ &= \frac{e^{\frac{2\pi i}{r}}}{\sqrt{r}} \left(e^{-\frac{2\pi i}{r}} U|\phi\rangle + e^{-2 \cdot \frac{2\pi i}{r}} U^2|\phi\rangle + \dots + e^{-r \cdot \frac{2\pi i}{r}} U^r|\phi\rangle \right) \\ &= e^{\frac{2\pi i}{r}} \cdot |\psi_1\rangle \end{aligned}$$

Order finding on a quantum computer

This generalizes to $|\psi_s\rangle$

$$|\psi_s\rangle = \frac{1}{\sqrt{r}} \left(|\phi\rangle + e^{-s \cdot \frac{2\pi i}{r}} U|\phi\rangle + e^{-2s \cdot \frac{2\pi i}{r}} U^2|\phi\rangle + \dots + e^{-(r-1)s \cdot \frac{2\pi i}{r}} U^{r-1}|\phi\rangle \right)$$

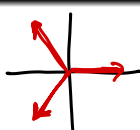
It has eigenvalue

$$U|\psi_s\rangle = e^{\frac{2\pi i s}{r}} |\psi_s\rangle$$

Idea: create *any* $|\psi_s\rangle$, run QPE to get estimate of s/r , compute r .

What's the problem?

Order finding on a quantum computer



$$e^{\frac{2\pi i}{3} \cdot s} \quad 1 + e^{-\frac{2\pi i}{r}} + e^{-\frac{2 \cdot 2\pi i}{r}} + \dots$$

$$e^{-(r-1) \cdot \frac{2\pi i}{r}}$$

Solution: construct the uniform superposition of all of them.

$$|\psi\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |\psi_s\rangle$$

But what does this equal?

$$|\psi_0\rangle = \frac{1}{\sqrt{r}} (|\phi\rangle + \cancel{0} |\phi\rangle + \cancel{0^2} |\phi\rangle + \dots)$$

$$|\psi_1\rangle = \frac{1}{\sqrt{r}} (|\phi\rangle + e^{\frac{2\pi i}{r}} \cancel{0} |\phi\rangle + \dots)$$

$$|\psi_2\rangle = \frac{1}{\sqrt{r}} (|\phi\rangle + e^{-2 \cdot \frac{2\pi i}{r}} \cancel{0} |\phi\rangle + \dots)$$

Order finding on a quantum computer

The superposition of all $|\Psi_s\rangle$ is just our original state $|\phi\rangle$!

$$\begin{aligned}
 |\Psi\rangle &= \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{k \cdot \frac{2\pi i}{N}} |\Psi_k\rangle = \frac{1}{\sqrt{r}} \left(|\Psi_0\rangle + |\Psi_1\rangle + \dots + |\Psi_{r-1}\rangle \right) \\
 &= \frac{1}{\sqrt{r}} \left(\frac{1}{\sqrt{r}} (|\phi\rangle + e^{\frac{-2\pi i}{r}} U|\phi\rangle + \dots + e^{\frac{-2\pi i(r-1)}{r}} U^{r-1}|\phi\rangle) \right. \\
 &\quad \left. + \frac{1}{\sqrt{r}} (|\phi\rangle + e^{\frac{-2\pi i(r-1)}{r}} U|\phi\rangle + \dots + e^{\frac{-2\pi i(r-1)^2}{r}} U^{r-1}|\phi\rangle) \right. \\
 &\quad \left. + \dots + \frac{1}{\sqrt{r}} (|\phi\rangle + e^{\frac{-2\pi i(r-1)}{r}} U|\phi\rangle + \dots + e^{\frac{-2\pi i(r-1)^2}{r}} U^{r-1}|\phi\rangle) \right) \\
 &\quad \quad \quad \downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow \\
 &\quad \quad \quad r \quad \quad \quad 0 \quad \quad \quad \dots \quad 0 \quad \quad \quad 0
 \end{aligned}$$

$$\begin{aligned}
 e^{s \cdot \frac{2\pi i}{4}} &: 1 + e^{\frac{\pi i}{2}} + e^{\pi i} + e^{\frac{3\pi i}{2}} \\
 &= 1 + i + (-1) + (-i) \\
 &= 0
 \end{aligned}$$

$$= \frac{1}{\sqrt{r}} \cdot \frac{1}{\sqrt{r}} \cdot r |\phi\rangle = |\phi\rangle = |1\rangle$$

If we run QPE, the output will be s/r for one of these states.

Image credit: Xanadu Quantum Codebook node S.3

Shor's algorithm: overview

Input: integer N , guaranteed to have structure

$$N = p \cdot q$$

for p, q prime.

Output: p and q

Factoring is accomplished by a **reduction** from order finding.

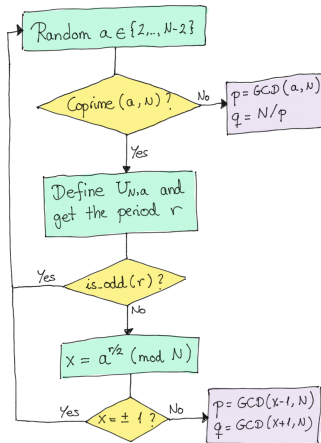


Image credit: Xanadu Quantum Codebook node S.3

Non-trivial square roots and factoring

Choose a random a that is coprime with N . If its order, r , is even,

$$a^r = a^{2r'} = (a^{r'})^2 \equiv 1 \pmod{N}$$

Let's rewrite and factor...

$$(a^{r'})^2 - 1 \equiv 0 \pmod{N}$$

$$(a^{r'} - 1)(a^{r'} + 1) \equiv 0 \pmod{N}$$

↙
 $s \cdot p$

↘
 $t \cdot q$

$$N = pq$$

Non-trivial square roots and factoring

This means

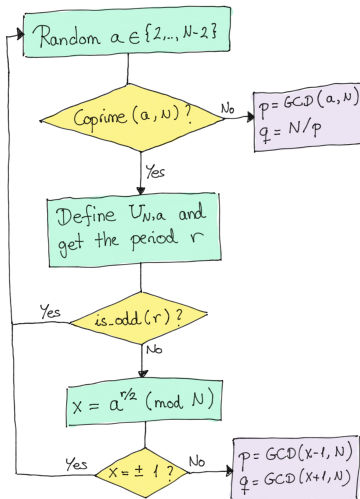
Based on structure of N ,

If $a^{r'} \neq \pm 1$, we can determine p and q

Is this really efficient?

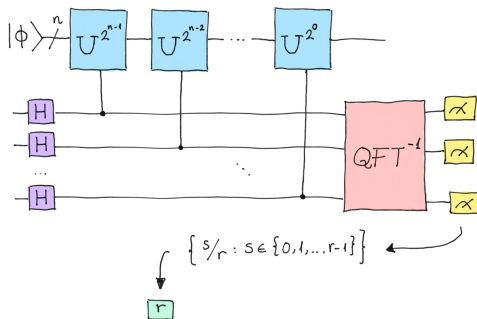
GCD: polynomial w/Euclid's algorithm

Modular exponentiation: can use exponentiation by squaring, other methods to reduce operations and memory required



Is this really efficient?

Quantum part: let $L = \lceil \log_2 N \rceil$.



QFT: gates polynomial in number of qubits $O(L^2)$

Controlled- U : requires $O(L^3)$ gates.

Discussion

Form pairs or small groups, and consider the following questions:

- 1 Shor's algorithm was developed in 1994. Estimate the fraction of today's world population that can actually implement it.
- 2 Shor's algo can be used to break cryptosystems like RSA. Estimate the proportion of the world that would be affected if someone actually deployed it at scale.
- 3 Is it ethical to develop such an algorithm? Is it ethical to *teach* such an algorithm?
- 4 Look up some resource estimates; how long would it actually take to break 2048-bit RSA? How many qubits are needed?
- 5 Think critically about (a) who knows how to implement the algorithm, and (b) who will potentially have access to quantum hardware in the future. What issues can you foresee?
- 6 What are ways we can keep our cryptographic infrastructure secure in the future?

Next time

Content:

- Tutorial 7: hands-on w/RSA and Shor's algorithm
- Module 4: quantum channels, and noise in quantum systems

Action items:

- 1 Tutorial assignment 3
- 2 Project midterm checkpoint report

Recommended reading:

- From this class: Codebook QFT, QPE, SH; N&C 5.3, A.5
- For next class: Codebook NT; N&C 8.1-8.3